POLI666 Replication Paper: Manual Replication Group

Leo Amorim, Alan Nemirovski, Aiden McIlvaney

April 9th, 2026

We went into this project with nothing but a dream and pure vibes…

Plan of attack

1. Defining the scope of AI use in our group
2. Translating from Stata to R
3. Building a codebook
4. Inspecting the data
5. Re-running original analysis
6. Evaluating the main DV + Alternative measures
7. Robustness checks
8. Next steps: Assessing other (potential) heterogeneous effects

AI assistance in our manual replication

Defining the limits of our AI use upfront:

- **No AI was used to write or generate any R code.** This process remained entirely manual.
- AI was only used for two specific purposes relating to the bridging of knowledge gaps that would be difficult to do otherwise:
  1. Compile a working survey codebook using partial variable labels
  2. Explain in plain language what a given line(s) of Stata code is attempting to execute, to inform how we should approach the replication in R

Building the (missing!) survey codebook

- No survey codebook is included in replication materials
- Dataset variable objects were named by question ID numbers
  - Labels contained original questions (in Russian), however…
- **Variable labels in Stata are truncated at 80 characters** → Questions were incomplete, particularly ones that repeated the same setup for different sub-items (e.g., q9a-q9j)
  - Used Claude to reconstruct a functional codebook – it was helpful in inferring some of the missing information!

Translating from Stata to R

- *RStata* package was explored as an option to execute Stata files directly in RStudio, but it still required installing the Stata language

- Original variable labels were in Russian and object names were assigned through question IDs

- Ended up relying on Claude (🙏)-built codebook and Appendix documents in the supplementary materials available on the University of Chicago Press Journals

```
*trust

gen trustarmy = q9a
recode trustarmy 99 =. 1=5 2=4 4=2 5=1

gen trustpolice = q9b
recode trustpolice 99=. 1=5 2=4 4=2 5=1

gen trustfsb = q9c
recode trustfsb 99=. 1=5 2=4 4=2 5=1

gen trustcourt = q9d
recode trustcourt 99=. 1=5 2=4 4=2 5=1

gen trustmungov = q9e
recode trustmungov 99=. 1=5 2=4 4=2 5=1
```

```
*Table 2

reg trustgovind period3 period2min protestday electday wealth education okrugdv1-okrugdv10, robust
outreg2 using tab2, keep(period3 period2min protestday electday) dec(2) alpha(0.001, 0.05) symbol(*,*) label re
reg trustarmy period3 period2min protestday electday wealth education okrugdv1-okrugdv10, robust
outreg2 using tab2, keep(period3 period2min protestday electday) dec(2)alpha(0.001, 0.05) symbol(*,*) label
reg trustpolice period3 period2min protestday electday wealth education okrugdv1-okrugdv10, robust
```

What Stata has that R does not

- Stata has certain functionalities that it executes automatically, which requires more explicit spelling-out in R:

  1. Automatically omits collinear predictor variables for analysis
     - The "okrug 10" (N = 29) has no United Russia respondents in the post-election window
     - This makes the UR X Post-Election interaction a constant that had to be manually fixed

  2. Different code format for some variable transformation
     - For example, creating individual dummies for each level of the *okrug* variable

  3. Missing (NA) values need to be handled more carefully/explicitly in R (e.g., requiring separate conditions to ignore NA values within functions like *ifelse*)

Inspecting the data

- The dataset appears consistent with the one used in the paper:
    - Total observations (N = 1,550) in the dataset match what is in the article
        - We found the number of interviews per day to be consistent with those listed in the Supplemental Appendix 1 table (pg. 7 of Supplemental Material)
    - Total variables = 292
    - There do not appear to be any duplicate IDs
    - There are no unrealistic values in the age range; all respondents are aged 18 to 85

Inspecting the data (cont'd)

- The results in the Appendix generally match what's in the paper
- Appendix 2A: the main difference is the N column for "Index" yields 1,521 compared to the paper's 1,232 (likely stems from Stata automatically dropping collinear items)
- Appendix 2B: essentially the same, with the only only difference stemming from the same N difference as before
- Appendices 3.1 and 3.2: the results are basically the same as in the paper; the Ns match the paper exactly

Inspecting the data (cont'd)

- Additionally, there are concerns relating to ***how*** the data was collected, as the dataset included in the replication files has many quality issues
  - Incompleteness
  - Difficulty identifying what some objects and variables refer to
  - Substantive amounts of data collected differentially across the population (e.g., partisanship questions, q62 only asked for 186 pre-election respondents)

- After further inspecting the data, there were two other key issues we think are worth highlighting:
  - The Federal Security Service (FSB) non-response rate triples from 4% to 15% post-protest; the United Nations non-response doubles from 11% to 24%
  - There are 318 (21%) missing observations from the initial trust index regression

Re-running the original analysis

	Gov Index	Army	Police	FSB	Court	Municipal Gov	Federal Gov	Duma	Procuracy	United Nations
Post-protest	0.13*	0.05	0.14*	0.17**	0.07	0.19***	0.21***	0.08	0.05	0.12+
	(0.05)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.07)
Post-election	0.11	0.19+	0.12	0.01	0.15	0.04	−0.02	0.08	0.18+	−0.11
	(0.09)	(0.10)	(0.11)	(0.10)	(0.11)	(0.11)	(0.11)	(0.10)	(0.11)	(0.11)
Protest day	0.22+	0.53***	0.35*	0.18	0.30*	0.20	0.07	0.03	0.15	0.00
	(0.13)	(0.14)	(0.14)	(0.16)	(0.15)	(0.16)	(0.16)	(0.15)	(0.15)	(0.16)
Election day	−0.05	−0.23	0.09	0.11	0.07	−0.19	−0.25+	−0.18	0.03	−0.43**
	(0.12)	(0.16)	(0.15)	(0.17)	(0.15)	(0.14)	(0.15)	(0.13)	(0.14)	(0.16)
Constant	3.17***	4.02***	3.06***	3.28***	2.97***	3.34***	3.49***	3.04***	2.96***	3.16***
	(0.16)	(0.19)	(0.18)	(0.19)	(0.18)	(0.19)	(0.20)	(0.19)	(0.20)	(0.22)
Num.Obs.	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R2	0.088	0.066	0.052	0.085	0.057	0.072	0.085	0.062	0.054	0.042

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Re-generated Table 2 from Frye and Borisova (2019)

- Differences in trust (Table 1) and coefficient estimates (Table 2) were slightly different at times, despite consistent sample sizes
    - However, differences don't appear to be substantially different - direction and statistical significance of the results remained the same

Evaluating the DV: Factor analysis

- Exploratory (EFA) and confirmatory (CFA) factor analysis was done on all eight variables, with similar factor loading scores
  - Using *psych* and *lavaan* packages in R, respectively
  - Cronbach's alpha ~ 0.903 (raw and standardized)

variable	efa.loading.score	cfa.loading.score
trustarmy	0.5296033	0.5288245
trustpolice	0.6906914	0.6932360
trustfsb	0.7031063	0.6874910
trustcourt	0.6796468	0.6942999
trustmungov	0.8200351	0.8422222
trustfedgov	0.8462218	0.8723872
trustduma	0.8168069	0.8414339
trustprocuracy	0.7175109	0.7246028

- Implication: Government trust index in the paper appears to capture the same underlying latent variable (trust in government)

Alternative measures of the DV

1. Including the Orthodox church in the trust index
   - It is the only institution (from q9 items) that was left out of the index
2. Dropping one factor at a time from the trust index
3. Sub-indices based on alternative concepts of trust (Schenider 2017)
4. Loading score-weighted trust index

All regression models with alternative trust indices yielded statistically significant coefficient on the *postprotest* variable (similar magnitudes) at the 5% significance level or higher

- Importantly, all variations of this index worked with items from q9a-q9j

Alternative measures of the DV: Results

Table 2: Accounting for the role of Church

	Gov Index (with Church)	Church
Post-protest	0.13**	0.10+
	(0.05)	(0.06)
Post-election	0.08	0.06
	(0.09)	(0.11)
Protest day	0.17	0.03
	(0.12)	(0.16)
Election day	−0.07	−0.30
	(0.12)	(0.20)
Num.Obs.	1192	1445
R2	0.087	0.042

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3: Robustness check on Gov't Trust: Leave-one-out indices

	No Army	No Police	No FSB	No Court	No Municipal Gov	No Federal Gov	No Duma	No Procuracy
Post-protest	0.14**	0.13**	0.11*	0.13**	0.12*	0.12*	0.12*	0.14**
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Post-election	0.08	0.09	0.12	0.10	0.13	0.14	0.13	0.08
	(0.10)	(0.09)	(0.08)	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)
Protest day	0.16	0.18	0.22+	0.20	0.24+	0.26*	0.27*	0.21
	(0.14)	(0.13)	(0.12)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)
Election day	−0.03	−0.07	−0.09	−0.07	−0.03	−0.03	−0.03	−0.05
	(0.12)	(0.12)	(0.12)	(0.12)	(0.12)	(0.12)	(0.12)	(0.12)
Num.Obs.	1242	1235	1332	1243	1235	1236	1242	1260
R2	0.082	0.088	0.078	0.089	0.090	0.089	0.088	0.087

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Alternative measures of the DV: Results (cont'd)

Table 4: Sub-indices constructed following Schneider (2017)

	Model 1	Model 2.1	Model 2.2	Model 3.1	Model 3.2
Post-protest	0.16**	0.12*	0.14**	0.13**	0.18**
	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)
Post-election	0.04	0.11	0.08	0.09	0.08
	(0.10)	(0.08)	(0.09)	(0.09)	(0.12)
Protest day	0.12	0.26*	0.23+	0.16	0.20
	(0.15)	(0.12)	(0.13)	(0.13)	(0.17)
Election day	−0.21	−0.14	−0.06	−0.06	−0.01
	(0.13)	(0.12)	(0.13)	(0.12)	(0.15)
Num.Obs.	1438	1417	1282	1388	1275
R2	0.079	0.085	0.094	0.070	0.082
Fed Gov	Yes	Yes	Yes	Yes	Yes
Duma	Yes	Yes	Yes	Yes	Yes
Mun Gov	Yes				
Police		Yes	Yes	Yes	Yes
Army		Yes	Yes		
Court				Yes	Yes
FSB			Yes		Yes

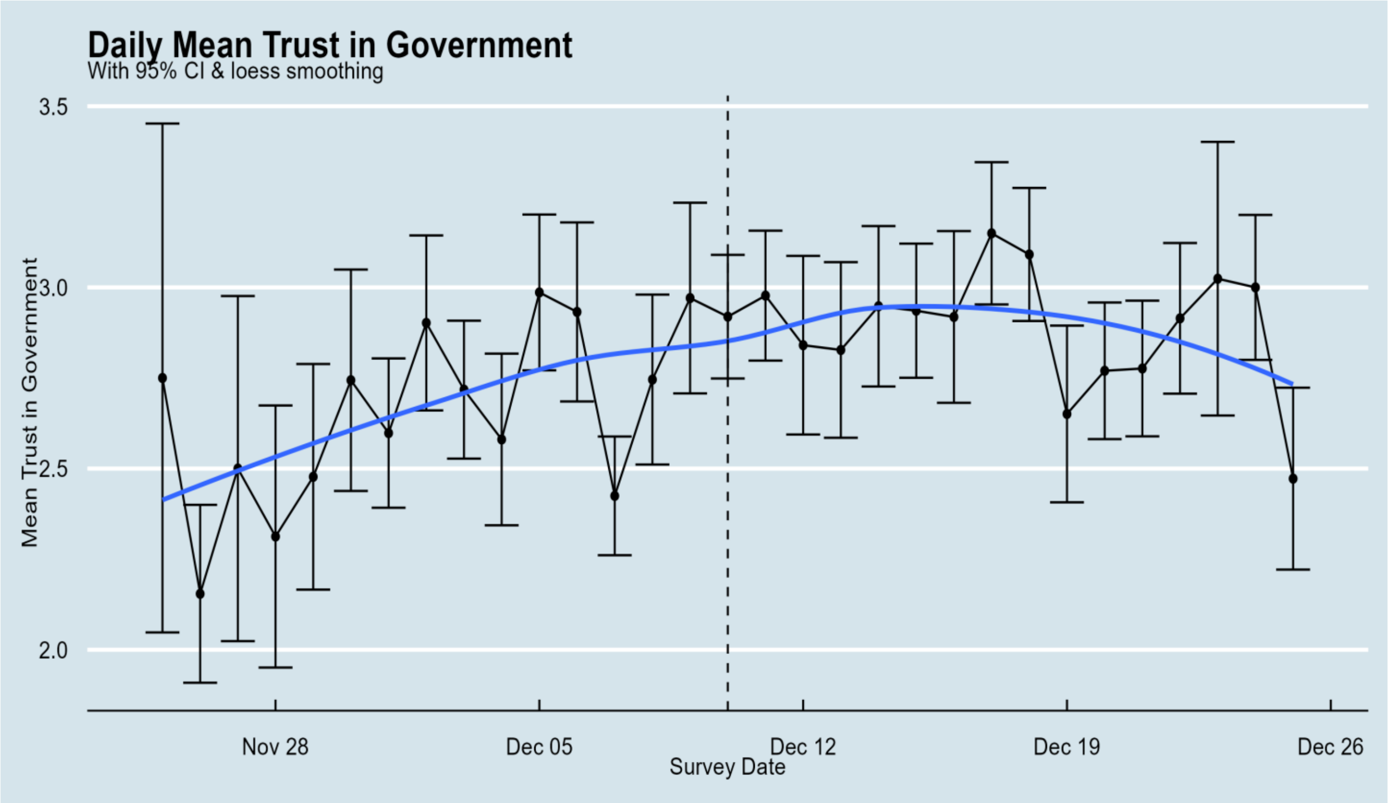
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Robustness check on Gov't Trust: Factor Loading Score-Weighted Index

	(1)
Post-protest	0.13**
	(0.05)
Post-election	0.09
	(0.09)
Protest day	0.19
	(0.13)
Election day	−0.05
	(0.12)
Num.Obs.	1232
R2	0.086

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Robustness checks: Placebo tests



- This paper treats Dec. 10 as a sharp treatment date
- Trust appears noisy and trending upward before the protest, making the validity of some assumptions shaky
- I would not want to be the authors right now…but this graph alone does not prove anything

Robustness checks: Placebo tests

- After re-running the main regression with a post-treatment indicator for each date in the table, the estimated effects turn out to be similar for Dec. 8, 9, and 10
- Since the effect appears before the treatment, the paper's assumptions of no pre-treatment effects and correct treatment timing seem shaky
- To be fair, there were smaller protests happening since Dec. 6 – but treating Dec. 10 as a sharp treatment date is flawed

Placebo Timing Tests

Post-treatment coefficients using alternative protest dates

Date	Post Coef.	p-value
Dec 06	0.067	0.202
Dec 07	0.077	0.142
Dec 08	0.148	0.003
Dec 09	0.146	0.002
Dec 10	0.140	0.003
Dec 11	0.117	0.013

Robustness checks: Manipulating controls

- Re-tested the model with no controls, baseline controls and additional covariates to check whether day-to-day differences in the sample affected outcomes.
  - Results are largely similar to original:
  - No controls: **0.254*** (bigger effect)
  - Baseline controls: **0.168**** (almost identical)
  - All controls: 0.085 (not significant)
- While there is some sensitivity to model specification, the results seem not to be affected by the constituents of the sample.

Robustness checks:

Non-response analysis and worst-case modelling

- Appendix IV. Non-responses:
  - FSB non-response rate jumps from 4% (pre-election) to 15% (post-protest)
  - UN non-response rate jumps from 11% (pre-election) to 24% (post-election)
- The problem: if the respondents who declined to rate the FSB and UN post-protest are disproportionately distrustful, then the observed post-protest trust increase is upwardly biased
- As a robustness measure, we rerun Appendix Table 2A under the assumption that ***all*** non-responders are maximally distrustful (the "nuclear" option)

AWM

Robustness checks:

(Original) Table SA-2A Determinants of Trust in Institutions

	Gov't Index 1	Army 2	Police 3	FSB 4	Court 5	City Gov't 6	Fed Gov't 7	Duma 8	Procuracy 9	UN 10
Post-Protest	.16*	.23*	.28*	.27*	.13	.17*	.17*	.05	-.01	.05
	(.06)	(.08)	(.07)	(.08)	(.07)	(.07)	(.08)	(.08)	(.08)	(.09)
Post-Election	.04	.28*	.24*	.18*	.08	-.04	-.05	-.06	-.14	-.11
	(.07)	(.09)	(.08)	(.09)	(.09)	(.08)	(.09)	(.09)	(.09)	(.10)
Protest day	.13	.50*	.38*	.28*	.20	.14	.07	-.09	-.12	.05
	(.10)	(.12)	(.11)	(.12)	(.11)	(.12)	(.12)	(.12)	(.12)	(.13)
Election day	-0.03	-0.09	0.22	0.21	0.11	-0.21	-0.28	-0.21	-0.04	-0.49*
	(0.12)	(0.16)	(0.16)	(0.17)	(0.15)	(0.15)	(0.15)	(0.14)	(0.15)	(0.17)
Constant	3.18	3.94	2.99	2.30	2.95	3.35	3.50	3.06	3.05	2.91
	(.17)	(.20)	(.18)	(.26)	(.18)	(.19)	(.20)	(.20)	(.25)	(.24)
F-test: Post-Protest = Post-Election	3.93	.65	.41	1.49	.45	10.1	11	2.95	3.46	3.93
Prob > F	.05	.4	.5	.2	.5	.001	.001	.09	.06	.05
Obs	1232	1491	1505	1338	1450	1485	1487	1466	1437	1212
R-squared	.09	.07	.06	.09	.06	.07	.09	.06	.05	.04

Non-response analysis: Results

- The results break down institutionally:
  - Army 0.227* → 0.188*
  - Police 0.285* → 0.275*
  - City Government 0.172* → 0.153*
  - FSB 0.273* → **0.076**
  - UN 0.5 → **–0.159**
- In sum, the FSB loses its statistical significance under the worst-case scenario

Robustness Check: Partial Response Trust Index

- In the original paper, there are 318 (21%) observations that are dropped from the trust index regression
- The concern here is that this attrition is not random
- We thus investigate whether the loss is driven by missing trust items or by missing control variables

Robustness Check: Results

- When the government trust index is recomputed as the mean of whichever items each respondent answered (rather than requiring all eight) zero additional observations are recovered relative to the main model
- The partial-response robustness check confirms that the 21% sample attrition in the trust index model is entirely driven by missing control variables rather than missing trust items

Next steps

- Assessing other potential heterogeneous + subgroup-level effects
  - Power analysis and balance tests on *all* (disaggregated) subgroups
- Looking (more descriptively) at the role of other potentially confounding events happening during the post-protest period
  - Protests occurring continuously, and daily trust levels remain highly volatile
  - Dec. 24: Start of second wave of mass protests
- Other ways to conceptualize trust? (as was somewhat done in SA3)

Summary of key insights (so far…)

- Manually replicating the original analysis in R yielded **slightly different coefficients, but of similar magnitudes**, as well as the **same directionality and statistical significance**

- **Post-survey methodology seems to MOSTLY check out**

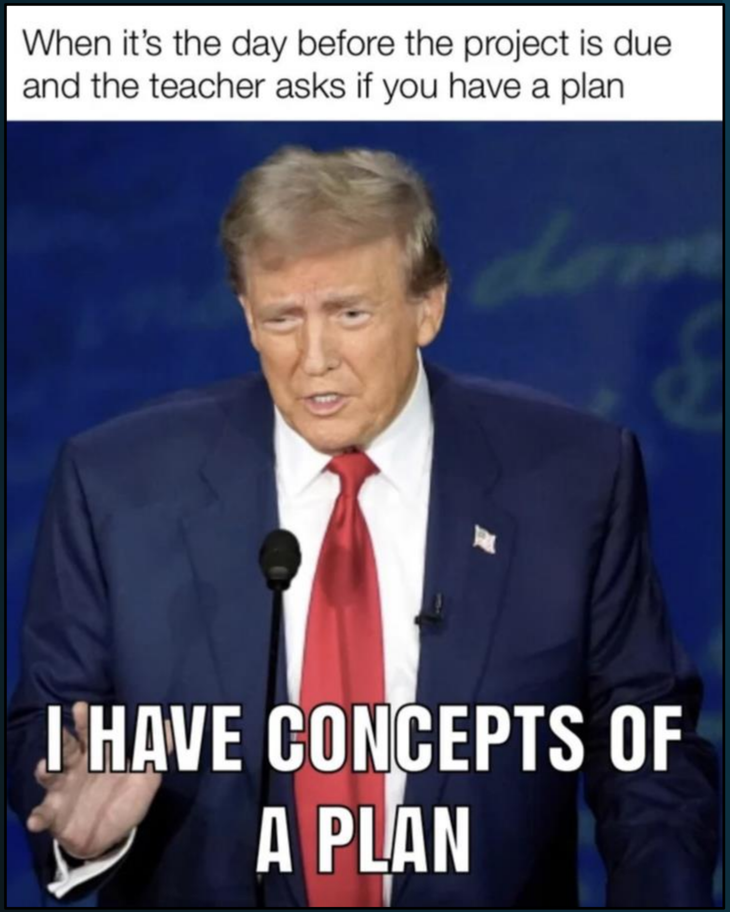
- **Main issues seem to be related to the quality of the data** and collection process
    - Some of this is presumably endemic to an UED study
    - Some of it is avoidable (e.g., data labelling, codebook, etc.)

- Ideally, there would be another study to assess the external validity of the paper

Plan for final report

Based on template report (Overleaf file):

1. Introduction
2. Defining parameters of the replication + scope of AI use
3. Translation process + main issues
   - Coding log
   - Environment documentation
4. Replicating original analysis
5. Further tests + robustness checks
6. Main takeaways + conclusion



Thank You!